

Augustus (Jiashang) Shen

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EDUCATION

University of Southern California — Dornsife College of Letters, Arts & Sciences

Los Angeles, CA

B.S. Quantitative Biology, Minor in Chemistry · Cumulative GPA 3.959 / 4.000 · Expected May 2028

- **Honors:** BUGS Research Award; Dornsife Dean's List; Phi Kappa Phi (Sophomore Academic Recognition); Academic Achievement Award
- **Scholarships:** SOAR Research Fund; SURF Research Fund
- **Coursework:** Cell Biology; General Physiology; Systems Biology; Analysis of Biological Data; Statistics; Programming in Python; Object-Oriented Programming; Discrete Methods in Computer Science; Organic & General Chemistry; Calculus

High School Affiliated to Shanghai Jiao Tong University

Shanghai, China

IB Diploma · SAT 1530 · June 2024

- **Honors:** Principal's Award; Outstanding Student of the Class of 2024; Excellent Student Leader Award
- **Coursework:** IB Biology HL, Chemistry HL, Mathematics AI HL; Chinese Literature SL, Economics SL, English B SL

RESEARCH EXPERIENCE

Foster Laboratory, University of Southern California

Jan 2025 – Present

Undergraduate Researcher — Quantitative Cell Biology & Non-equilibrium Physics

- Investigate the physical bioenergetics of active cytoskeletal systems and the emergence of organization in non-equilibrium biological materials.
- Develop and maintain molecular-biology, protein-expression, and in vitro reconstitution workflows for kinesin–microtubule research.
- Contribute to experimental design and protocol optimization spanning cloning, baculovirus expression, protein purification, and microscopy-based assays.
- Trained in spinning-disk confocal and LC-PolScope microscopy and quantitative imaging of cytoskeletal organization and dynamics.
- Present research progress at lab meetings and an undergraduate research symposium.

RESEARCH PROJECTS

Energy Dissipation & Self-Organization in Kinesin–Microtubule Active Networks

Apr 2025 – Present

Lead Undergraduate Researcher — Foster Laboratory

- Lead a project on how ATP-driven molecular motors generate emergent structural order in active microtubule networks.
- Independently developed the maize kinesin-14 system: sequence design, plasmid construction, cloning optimization, sequence verification, and bacmid generation for insect-cell expression.
- Completed bacmid preparation and initiated insect-cell expression of mCherry-tagged maize kinesin-14 (one of five phylogenetically diverse homologs in a comparative study).

- Performing biophysical characterization: ATPase activity, microtubule-binding and active-contraction assays, and motor-driven network reconstitution.
- Integrate NADH-coupled fluorescence assays with LC-PolScope microscopy to relate motor energy dissipation to emergent microtubule alignment.

Modeling Plus-End Accumulation & Traffic Jams of Kinesin Motors (QBIO 482)

2025

- Built a stochastic 1D-lattice model of kinesin transport with binding, stepping, steric exclusion, and site-specific dissociation kinetics.
- Implemented the Gillespie stochastic simulation algorithm (SSA) in Python to study emergent traffic-jam formation under varying motor concentration and end-release rates.
- Identified plus-end dissociation kinetics as the dominant regulator of accumulation, and a non-monotonic concentration–efficiency relationship (crowding-induced jamming).
- Validated results against Leduc et al. (PNAS, 2012), linking transport dynamics to axonal-transport dysfunction in neurodegeneration.

LEADERSHIP

BUGS (Bridge Undergraduate Science Program) — Research Presenter

May – Aug 2025

Delivered oral presentations on independent kinesin-14 research; communicated design, failure analysis, and future directions to interdisciplinary audiences.

FuSSO Shanghai (Science Outreach Organization) — Founder / President

Jun 2023 – Aug 2025

Founded a student-led outreach initiative providing lab exposure to under-resourced high-school students; coordinated mentorship and partnered with local labs and biotech organizations.

Mendel's Pea (Biology Club) — Founder / President

Mar 2022 – Jun 2024

Led weekly biology sessions for 40+ members; recognized as a national pioneer / four-star club; earned 20+ medals across biology research and knowledge competitions.

SKILLS

Molecular Biology: Plasmid design & cloning, PCR, gel electrophoresis, bacterial & bacmid transformation, bacmid purification, DNA/RNA purification, baculovirus expression, protein purification.

Imaging: Spinning-disk confocal microscopy; LC-PolScope microscopy; quantitative image analysis.

Computation: Python (NumPy, pandas, Matplotlib, scikit-learn), stochastic simulation (Gillespie/SSA), R, C++.